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Supplementary material for Fed-GEE: Federated Generalized Estimating Equations for Privacy-Preserving Population-Level Inference for Multicenter Longitudinal Data

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1. Supplement A: Algorithm for inference and estimation for Fed-GEE

Require: Distributed datasets $\{\mathbf{X}_i, \mathbf{Y}_i\}_{i=1}^K$ across K sites; Initial parameter vector $\boldsymbol{\beta}^{(0)}$ (with site-level covariates initialized to 0); Working correlation parameter α; Convergence tolerance ϵ; Maximum iterations M.

Ensure: Estimated marginal parameters $\hat{\boldsymbol{\beta}}$; KC-corrected robust sandwich variance $\mathbf{V}_\beta^{KC}$.

Phase 1: Federated Fisher Scoring: Iterative optimization
for $m = 0, 1, \ldots, M-1$ **do**
  Server broadcasts global parameters $\boldsymbol{\beta}^{(m)}$ to all K sites.
  for each site $i \in \{1, \ldots, K\}$ **in parallel do**
    // Execute locally without exposing patient data
    Compute local Score vector: $\mathbf{S}_i\left(\boldsymbol{\beta}^{(m)}\right) = \sum_{j=1}^{n_i} \mathbf{D}_{ij}^T \mathbf{V}_{ij}^{-1}\left(\mathbf{Y}_{ij} - \boldsymbol{\mu}_{ij}\right)$
    Compute local Bread matrix: $\mathbf{B}_i\left(\boldsymbol{\beta}^{(m)}\right) = \sum_{j=1}^{n_i} \mathbf{D}_{ij}^T \mathbf{V}_{ij}^{-1} \mathbf{D}_{ij}$
    Transmit $\{\mathbf{S}_i, \mathbf{B}_i\}$ to the central server.
  end for
  Server Execution:
  Aggregate global matrices: $\mathbf{S}_{global} = \sum_{i=1}^K \mathbf{S}_i$ and $\mathbf{B}_{global} = \sum_{i=1}^K \mathbf{B}_i$
  Update parameters: $\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \mathbf{B}_{global}^{-1}\mathbf{S}_{global}$
  if $\left|\left|\boldsymbol{\beta}^{(m+1)} - \boldsymbol{\beta}^{(m)}\right|\right| < \epsilon$ **then**
    Set $\hat{\boldsymbol{\beta}} \leftarrow \boldsymbol{\beta}^{(m+1)}$ and **break**
  end if
end for
Phase 2: Kauermann-Carroll Small-Sample Variance Estimation
Server broadcasts final parameters $\hat{\boldsymbol{\beta}}$ and final global Bread $\mathbf{B}_{global}$ to all K sites.
for each site $i \in \{1, \ldots, K\}$ **in parallel do**
  Compute final local Score evaluated at convergence: $\mathbf{S}_i\left(\hat{\boldsymbol{\beta}}\right)$
  Compute local Bread evaluated at convergence: $\mathbf{B}_i\left(\hat{\boldsymbol{\beta}}\right)$
  Calculate local site leverage: $\mathbf{H}_{ii} = \mathbf{B}_i \mathbf{B}_{global}^{-1}$
  Apply leverage to adjust score: $\widetilde{\mathbf{S}}_i = \left(\mathbf{I} - \mathbf{H}_{ii}\right)^{-1/2} \mathbf{S}_i\left(\hat{\boldsymbol{\beta}}\right)$
  Construct bias-corrected local Meat matrix: $\mathbf{M}_i^{KC} = \widetilde{\mathbf{S}}_i \widetilde{\mathbf{S}}_i^T$
  Transmit $\mathbf{M}_i^{KC}$ to the central server.
end for
Server Execution:
Aggregate global corrected Meat matrix: $\mathbf{M}_{global}^{KC} = \sum_{i=1}^K \mathbf{M}_i^{KC}$
Construct final robust variance: $\mathbf{V}_\beta^{KC} = \mathbf{B}_{global}^{-1} \mathbf{M}_{global}^{KC} \mathbf{B}_{global}^{-1}$
return $\hat{\boldsymbol{\beta}}$ and $\mathbf{V}_\beta^{KC}$

2. Supplement B: Proof of Theorem 1

Proof. Let $\mathbf{S}_{cen}(\boldsymbol{\beta})$ and $\mathbf{B}_{cen}(\boldsymbol{\beta})$ represent the score vector and information (bread) matrix for the centralized dataset. By the additive property of matrix operations over independent subjects across sites, we can decompose the centralized matrices:

$$\mathbf{S}_{cen}(\boldsymbol{\beta}) = \sum_{i=1}^{K}\sum_{j=1}^{n_i} \mathbf{D}_{ij}^T \mathbf{V}_{ij}^{-1}(\mathbf{Y}_{ij} - \boldsymbol{\mu}_{ij}) = \sum_{i=1}^{K} \mathbf{S}_i(\boldsymbol{\beta}) = \mathbf{S}_{global}(\boldsymbol{\beta}) \tag{1}$$

$$\mathbf{B}_{cen}(\boldsymbol{\beta}) = \sum_{i=1}^{K}\sum_{j=1}^{n_i} \mathbf{D}_{ij}^T \mathbf{V}_{ij}^{-1}\mathbf{D}_{ij} = \sum_{i=1}^{K} \mathbf{B}_i(\boldsymbol{\beta}) = \mathbf{B}_{global}(\boldsymbol{\beta}) \tag{2}$$

At any iteration m, assuming $\boldsymbol{\beta}_{fed}^{(m)} = \boldsymbol{\beta}_{cen}^{(m)}$, the centralized Newton-Raphson update is $\boldsymbol{\beta}_{cen}^{(m+1)} = \boldsymbol{\beta}_{cen}^{(m)} + \mathbf{B}_{cen}^{-1}\mathbf{S}_{cen}$. Substituting the sums yields exactly the Fed-GEE server update rule defined in Equation 1. By induction, the iterative sequences are identical, yielding $\hat{\boldsymbol{\beta}}_{fed} = \hat{\boldsymbol{\beta}}_{cen}$.

3. Supplement C: Proof of Theorem 2

Proof. By Theorem 1, $\hat{\boldsymbol{\beta}}_{fed}$ is the root of the global score function such that $\mathbf{S}_{global}(\hat{\boldsymbol{\beta}}_{fed}) = \mathbf{0}$. We take a first-order Taylor series expansion of $\mathbf{S}_{global}(\hat{\boldsymbol{\beta}}_{fed})$ around the true $\boldsymbol{\beta}_0$:

$$\mathbf{0} = \mathbf{S}_{global}\left(\boldsymbol{\beta}_0\right) + \left[\frac{\partial \mathbf{S}_{global}\left(\boldsymbol{\beta}^*\right)}{\partial \boldsymbol{\beta}^T}\right]\left(\hat{\boldsymbol{\beta}}_{fed} - \boldsymbol{\beta}_0\right) \tag{3}$$

where $\boldsymbol{\beta}^*$ lies on the line segment between $\hat{\boldsymbol{\beta}}_{fed}$ and $\boldsymbol{\beta}_0$. Note that for generalized estimating equations, the negative derivative of the score function is approximated by the bread matrix, $-\frac{\partial \mathbf{S}_{global}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^T} \approx \mathbf{B}_{global}(\boldsymbol{\beta})$. Rearranging the terms and multiplying by $\sqrt{K}/K$, we obtain:

$$\sqrt{K}\left(\hat{\boldsymbol{\beta}}_{fed} - \boldsymbol{\beta}_0\right) = \left(\frac{1}{K}\mathbf{B}_{global}(\boldsymbol{\beta}^*)\right)^{-1}\left(\frac{1}{\sqrt{K}}\mathbf{S}_{global}(\boldsymbol{\beta}_0)\right) \tag{4}$$

We analyze the two components on the right-hand side separately. First, because the marginal mean is correctly specified (assumption 1), we have $E[\mathbf{S}_i(\boldsymbol{\beta}_0)] = \mathbf{0}$. Furthermore, bounded cluster sizes (assumption 2) and bounded covariates (assumption 3) ensure that the variance of each independent site-level score vector, is strictly finite. The global score evaluated at $\boldsymbol{\beta}_0$ is a sum of K independent, zero-mean random vectors. By the multivariate

central limit theorem, we have:

$$\frac{1}{\sqrt{K}}\mathbf{S}_{global}(\boldsymbol{\beta}_0) = \frac{1}{\sqrt{K}}\sum_{i=1}^{K}\mathbf{S}_i(\boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{M}(\boldsymbol{\beta}_0)) \tag{5}$$

Second, the application of the weak law of large numbers requires the expected value of the local bread matrices to be bounded. Because the local bread matrix is a function of the covariates and is summed over the cluster size K assumption 2 and 3 guarantee that the elements remain finite. Consequently, under assumption 4, the weak law of large numbers implies that $\frac{1}{K}\mathbf{B}_{global}(\boldsymbol{\beta}^*) \xrightarrow{p} \mathcal{B}(\boldsymbol{\beta}_0)$. Since $\mathcal{B}(\boldsymbol{\beta}_0)$ is invertible, by the continuous mapping theorem, the inverse converges in probability to $\mathcal{B}^{-1}(\boldsymbol{\beta}_0)$. Combining these results using Slutsky's theorem, we establish asymptotic normality:

$$\sqrt{K}(\hat{\boldsymbol{\beta}}_{fed} - \boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{B}^{-1}(\boldsymbol{\beta}_0) \cdot \mathcal{N}(\mathbf{0}, \mathcal{M}(\boldsymbol{\beta}_0)) \equiv \mathcal{N}\left(\mathbf{0}, \mathcal{B}^{-1}\mathcal{M}\mathcal{B}^{-1}\right) \tag{6}$$

Consistency (i.e., $\hat{\boldsymbol{\beta}}_{fed} \xrightarrow{p} \boldsymbol{\beta}_0$) follows immediately from this asymptotic normality.

Finally, for the Kauermann-Carroll adjusted meat matrix $\mathbf{M}^{KC}_{global}$, observe that the local leverage matrices are defined as $\mathbf{H}_{ii} = \mathbf{B}_i\left(\sum_i \mathbf{B}_i\right)^{-1}$. As $K \to \infty$, the global information matrix grows unboundedly, pushing the leverage terms $\mathbf{H}_{ii} \to \mathbf{0}$. Consequently, the adjustment term $(\mathbf{I} - \mathbf{H}_{ii})^{-1/2} \to \mathbf{I}$, demonstrating that the KC correction vanishes asymptotically, yielding the standard consistent sandwich estimator while providing necessary Type I error control in finite-sample distributed settings.

Remark 1 (Score-Level Leverage Correction): We note that the classical Kauermann-Carroll correction operates on the observation-level residuals via the $N_i \times N_i$ hat matrix $\mathbf{H}_i = \mathbf{D}_i\mathbf{B}^{-1}\mathbf{D}_i^T\mathbf{V}_i^{-1}$. In our federated setting, transmitting observation-level adjusted residuals would compromise the privacy guarantees of the protocol. We therefore apply an analogous correction in the score space, replacing the observation-level leverage with the $p \times p$ score-level leverage $\mathbf{H}_{ii}^{(s)} = \mathbf{B}_i\mathbf{B}^{-1}$.

[Table 1 about here.]

Table 1

Baseline characteristics of Veterans with Alcohol Use Disorder (N=29,041) stratified by MAUD initiation following inpatient discharge across 118 VHA facilities.

		MAUD initiation following inpatient discharge		
		No	*Yes*	Total
	N (%)	20,109 (69.2%)	8,932 (30.8%)	29,041 (100.0%)
Demographics	**Age Category**			
	18-39	3,920 (19.5%)	2,154 (24.1%)	6,074 (20.9%)
	40-64	11,411 (56.7%)	5,193 (58.1%)	16,604 (57.2%)
	65+	4,778 (23.8%)	1,585 (17.7%)	6,363 (21.9%)
	Sex			
	Male	19,083 (94.9%)	8,323 (93.2%)	27,406 (94.4%)
	Female	1,026 (5.1%)	609 (6.8%)	1,635 (5.6%)
	Race and Ethnicity			
	Black or African American	3,477 (17.3%)	1,486 (16.6%)	4,963 (17.1%)
	Hispanic	1,411 (7.0%)	627 (7.0%)	2,038 (7.0%)
	Missing/Other	1,364 (6.8%)	684 (7.7%)	2,048 (7.1%)
	White	13,150 (65.4%)	5,865 (65.7%)	19,015 (65.5%)
	Rurality			
	Missing	162 (0.8%)	59 (0.7%)	221 (0.8%)
	Urban	10,998 (54.7%)	4,733 (53.0%)	15,731 (54.2%)
	Rural	8,949 (44.5%)	4,140 (46.4%)	13,089 (45.1%)
	Area Deprivation Index			
	Missing	1,071 (5.3%)	391 (4.4%)	1,462 (5.0%)
	1-25	2,762 (13.7%)	1,283 (14.4%)	4,045 (13.9%)
	26-50	5,112 (25.4%)	2,505 (28.0%)	7,617 (26.2%)
	51-75	5,769 (28.7%)	2,634 (29.5%)	8,403 (28.9%)
	76-100	5,395 (26.8%)	2,119 (23.7%)	7,514 (25.9%)
Substance Use History	**Any Prior-Year MAUD Use**			
	No	15,371 (76.4%)	6,189 (69.3%)	21,560 (74.2%)
	Yes	4,738 (23.6%)	2,743 (30.7%)	7,481 (25.8%)
	OUD Diagnosis			
	No	16,590 (82.5%)	7,901 (88.5%)	24,491 (84.3%)
	Yes	3,519 (17.5%)	1,031 (11.5%)	4,550 (15.7%)
	AUDIT-C Category			
	No Use	2,099 (10.4%)	894 (10.0%)	2,993 (10.3%)
	Low use	2,203 (11.0%)	871 (9.8%)	3,074 (10.6%)
	Unhealthy Use	2,633 (13.1%)	1,169 (13.1%)	3,802 (13.1%)
	High Risk Use	1,612 (8.0%)	697 (7.8%)	2,309 (8.0%)
	Very High Risk Use	8,238 (41.0%)	3,277 (36.7%)	11,515 (39.7%)
	Missing	3,324 (16.5%)	2,024 (22.7%)	5,348 (18.4%)
Medical Complexity	**VA CAN Score**			
	Missing	1,573 (7.8%)	962 (10.8%)	2,535 (8.7%)
	0-19	1,050 (5.2%)	734 (8.2%)	1,784 (6.1%)
	20-39	3,187 (15.8%)	1,940 (21.7%)	5,127 (17.7%)
	40-59	4,121 (20.5%)	1,889 (21.1%)	6,010 (20.7%)
	60-79	4,549 (22.6%)	1,699 (19.0%)	6,248 (21.5%)
	80-99	5,629 (28.0%)	1,708 (19.1%)	7,337 (25.3%)
	Frailty Category			
	Not/Pre Frail	11,052 (55.0%)	6,086 (68.1%)	17,138 (59.0%)
	Mild/Moderate Frail	6,704 (33.3%)	2,290 (25.6%)	8,994 (31.0%)
	Severely Frail	2,353 (11.7%)	556 (6.2%)	2,909 (10.0%)
Hospitalization Factors	**Discharging Speciality**			
	Psychiatry	7,154 (35.6%)	4,345 (48.6%)	11,499 (39.6%)
	Medicine	11,993 (59.6%)	4,247 (47.5%)	16,240 (55.9%)
	Discharge Disposition			
	Routine	17,409 (86.6%)	8,475 (94.9%)	25,884 (89.1%)
	Domicillary	52 (0.3%)	30 (0.3%)	82 (0.3%)
	Addiction Consult			
	None	16,851 (83.8%)	7,002 (78.4%)	23,853 (82.1%)
	Non-clinical Consult	2,761 (13.7%)	1,593 (17.8%)	4,354 (15.0%)
	Clinical Consult	497 (2.5%)	337 (3.8%)	834 (2.9%)